Feature Selection with Annealing

By

Charles Silkin

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PROJECT INSTRUCTIONS

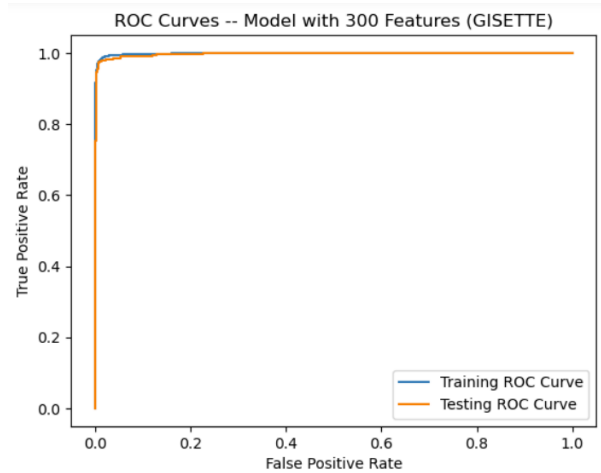
Implement the FSA variable selection method for linear models and binary classification with the logistic loss, as described in the slides (provided in folder). Use the parameters $s = 0.001$, $\mu = 100$, $N^{iter} = 300$. Take special care to **normalize each column** of the X matrix to have zero mean and variance 1 and to use the same mean and standard deviation that you used for normalizing the train set also for normalizing the test set.

a) Using the `Gisette` data, train a FSA classifier on the training set, starting with $\beta(0) = 0$ to select $k \in \{10, 30, 100, 300, 500\}$ features. Plot the training loss vs iteration number for $k = 300$. Report in a table the misclassification errors on the training and test set for the models obtained for all these k. Plot the misclassification error on the training and test set vs k. Also plot the train and test ROC curves of the obtained model with 300 features.

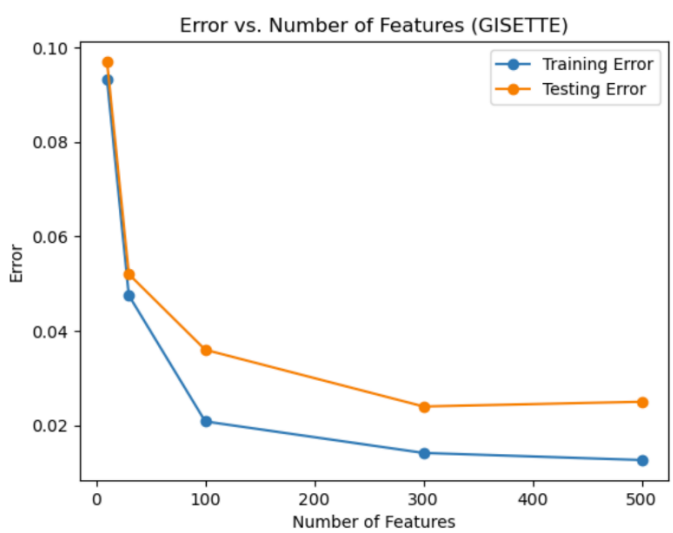
b) Repeat point a) on the `dexter` dataset.

c) Repeat point a) on the `madelon` dataset.

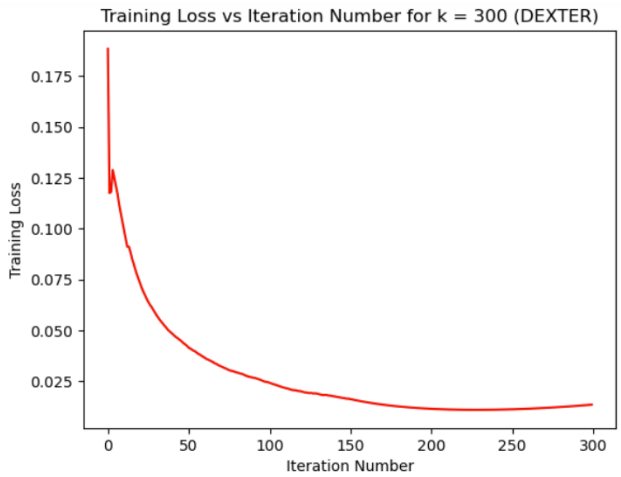
PART A

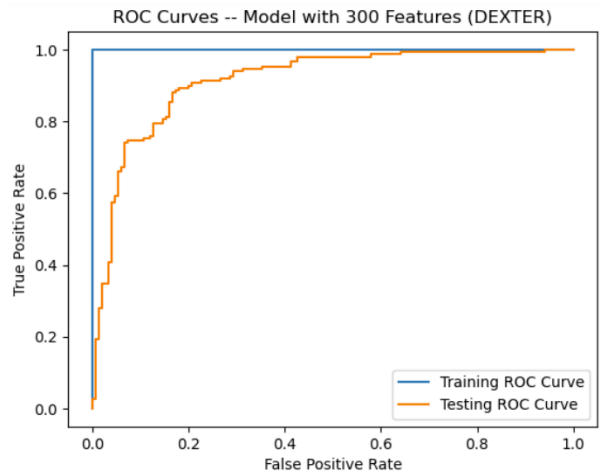


Number of Features	Training Error	Test Error
10	0.0933	0.097
30	0.0474	0.052
100	0.0208	0.036
300	0.0142	0.024
500	0.0127	0.025

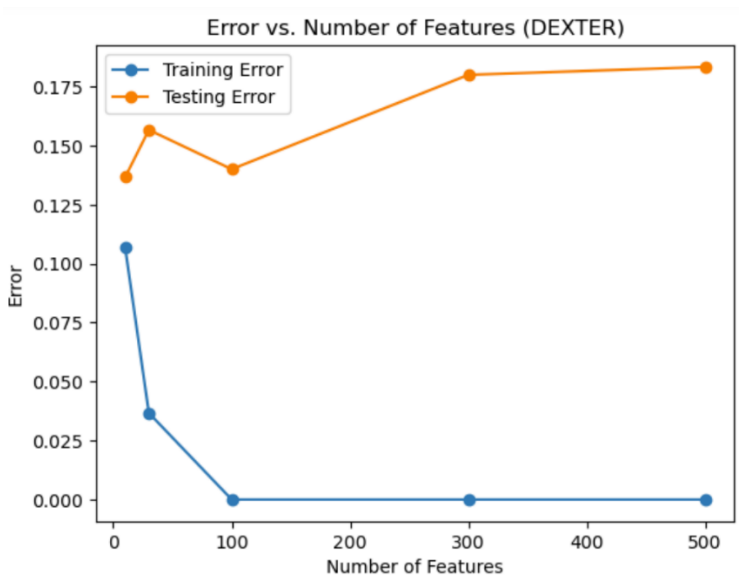


PART B

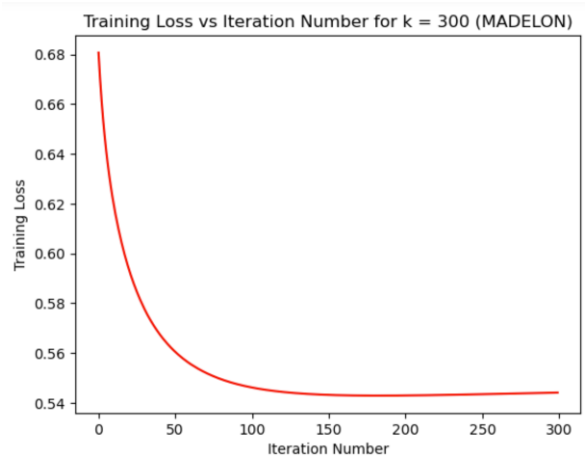


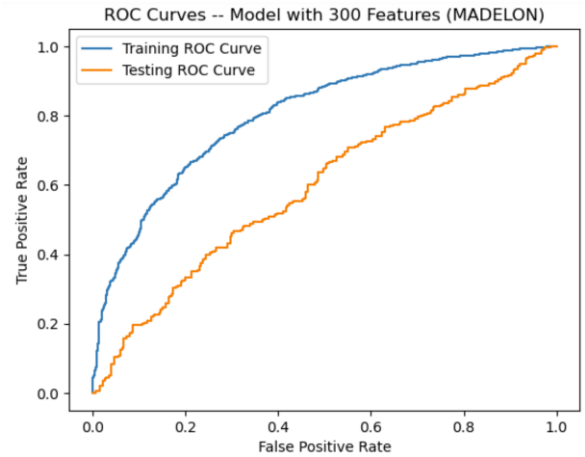


Number of Features	Training Error	Test Error
10	0.1067	0.1367
30	0.0367	0.1567
100	0.0	0.14
300	0.0	0.18
500	0.0	0.1833

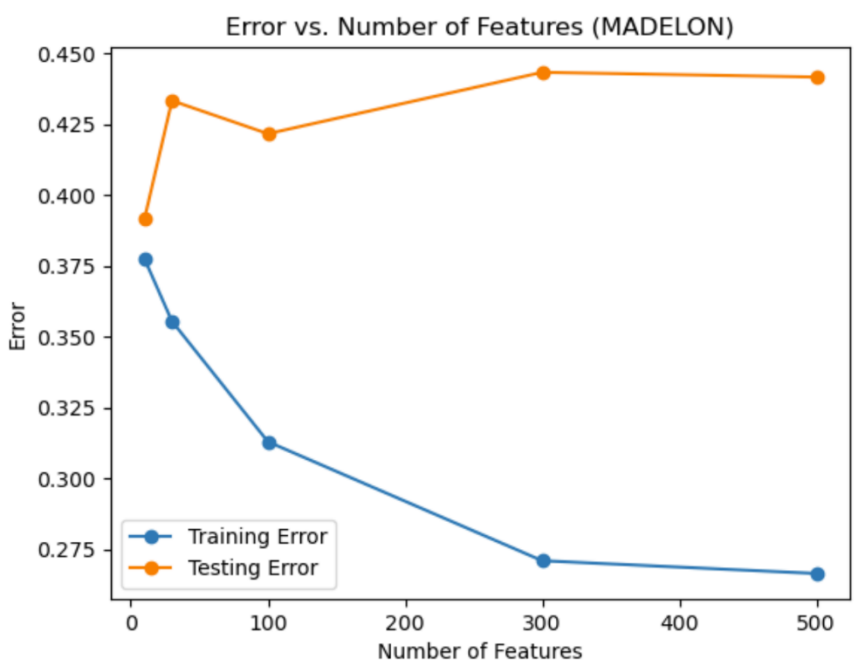


PART C





Number of Features	Training Error	Test Error
10	0.3775	0.392
30	0.3555	0.433
100	0.313	0.422
300	0.271	0.443
500	0.2665	0.442



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